

1 Methods

1.1 Core Idea

1.1.1 Setup

Consider a logistic regression, with missing features in the observation. Let $x := (x^o, x^m) \in \mathbb{R}^d$. Then the logistic regression is defined as

$$p(y \mid x) := \sigma(w^T x), \quad w \in \mathbb{R}^d.$$

Additionally we have a PC that models the density $p(x)$.

1.1.2 Goal

Compute the prediction in a probabilistically principled manner using

$$p(y \mid x^o) := \mathbb{E}_{p(x^m \mid x^o)} \left[\sigma(w^T x) \right].$$

If it were not for the non-linear sigmoid function σ , we would be able to calculate this in a smooth and decomposable PC, by pulling the Expectation -which is linear- into the leaf nodes. To tackle this we propose to use two tricks.

1.1.3 Trick 1: Pólya-Gamma

To replace the sigmoid function we use the PG-Sigmoid identity, shown in [1], that states:

$$\sigma(z) = \frac{1}{2} \exp\left(\frac{1}{2}z\right) \mathbb{E}_{\gamma \sim \text{PG}(1,0)} \left[\exp\left(-\frac{1}{2}\gamma z^2\right) \right]$$

for $z \in \mathbb{R}$.

If we set $z := w^T x$, and insert into our expectation, we get:

$$\begin{aligned} p(y \mid x^o) &= \mathbb{E}_{p(x^m \mid x^o)} [\sigma(z)] \\ &= \mathbb{E}_{p(x^m \mid x^o)} \left[\frac{1}{2} \exp\left(\frac{1}{2}z\right) \mathbb{E}_{\gamma \sim \text{PG}(1,0)} \left[\exp\left(-\frac{1}{2}\gamma z^2\right) \right] \right] \\ &= \frac{1}{2} \mathbb{E}_{p(x^m \mid x^o)} \mathbb{E}_{\gamma \sim \text{PG}(1,0)} \left[\exp\left(\frac{1}{2}z - \frac{1}{2}\gamma z^2\right) \right] \end{aligned}$$

Now, we have a term without a sigmoid, but the x still appears in a quadratic form.

1.1.4 Trick 2: Hubbard-Stratonovich (HS)

To get rid of the quadratic form, we apply the Hubbard–Stratonovich identity [2, 3], that states:

$$\exp\left(-\frac{1}{2}ah^2\right) = \sqrt{\frac{1}{2\pi a}} \int_{-\infty}^{\infty} \exp\left(-\frac{u^2}{2a} - ihu\right) du, \quad (1.1)$$

In plain words, it makes a quadratic term linear at the cost of introducing a helper variable that we need to integrate over.

Currently, in our term, x appears quadratically as well as linearly. To be able to apply HS, we complete the square (again using $z = w^T x$):

$$\begin{aligned} \frac{1}{2}z - \frac{1}{2}\gamma z^2 &= -\frac{1}{2}\gamma \left(z^2 - \frac{z}{\gamma} \right) \\ &= -\frac{1}{2}\gamma \left(\left(z - \frac{1}{2\gamma} \right)^2 - \frac{1}{4\gamma^2} \right) \\ &= -\frac{1}{2}\gamma \left(z - \frac{1}{2\gamma} \right)^2 + \frac{1}{2}\gamma \cdot \frac{1}{4\gamma^2} \\ &= -\frac{1}{2}\gamma \left(z - \frac{1}{2\gamma} \right)^2 + \frac{1}{8\gamma}. \end{aligned} \quad (1.2)$$

Now we apply HS with $a = \gamma$ and $h = z - \frac{1}{2\gamma}$:

$$\begin{aligned}
\exp\left(\frac{1}{2}z - \frac{1}{2}\gamma z^2\right) &= \exp\left(\frac{1}{8\gamma}\right) \exp\left(-\frac{1}{2}\gamma\left(z - \frac{1}{2\gamma}\right)^2\right) \\
&= \exp\left(\frac{1}{8\gamma}\right) \sqrt{\frac{1}{2\pi\gamma}} \int_{-\infty}^{\infty} \exp\left(-\frac{u^2}{2\gamma} - i\left(z - \frac{1}{2\gamma}\right)u\right) du \\
&= \exp\left(\frac{1}{8\gamma}\right) \int_{-\infty}^{\infty} \underbrace{\frac{1}{\sqrt{2\pi\gamma}} \exp\left(-\frac{u^2}{2\gamma}\right)}_{\mathcal{N}(u;0,\gamma)} \exp\left(-i\left(z - \frac{1}{2\gamma}\right)u\right) du \\
&= \exp\left(\frac{1}{8\gamma}\right) \mathbb{E}_{u \sim \mathcal{N}(0,\gamma)} \left[\exp\left(-i\left(z - \frac{1}{2\gamma}\right)u\right) \right] \\
&= \exp\left(\frac{1}{8\gamma}\right) \mathbb{E}_{u \sim \mathcal{N}(0,\gamma)} \left[\exp\left(\frac{i u}{2\gamma} - i z u\right) \right] \tag{1.3}
\end{aligned}$$

Substituting back into $p(y \mid x^o)$, we get:

$$\begin{aligned}
p(y \mid x^o) &= \frac{1}{2} \mathbb{E}_{p(x^m \mid x^o)} \mathbb{E}_{\gamma \sim \text{PG}(1,0)} \left[\exp\left(\frac{1}{2}z - \frac{1}{2}\gamma z^2\right) \right] \\
&= \frac{1}{2} \mathbb{E}_{p(x^m \mid x^o)} \mathbb{E}_{\gamma \sim \text{PG}(1,0)} \left[\exp\left(\frac{1}{8\gamma}\right) \mathbb{E}_{u \sim \mathcal{N}(0,\gamma)} \left[\exp\left(\frac{i u}{2\gamma} - i z u\right) \right] \right] \\
&= \frac{1}{2} \mathbb{E}_{\gamma \sim \text{PG}(1,0)} \mathbb{E}_{u \sim \mathcal{N}(0,\gamma)} \left[\exp\left(\frac{1 + 4i u}{8\gamma}\right) \mathbb{E}_{p(x^m \mid x^o)} [\exp(-i z u)] \right]
\end{aligned}$$

1.1.5 Characteristic Function

The characteristic function of a random variable X is defined as $\phi_X(s) = \mathbb{E}[\exp(isX)]$. The innermost expectation is only over x^m , with x^o fixed, so we split $z = w^T x$ into its observed and missing parts:

$$z = w^T x = w_o^T x^o + w_m^T x^m.$$

The x^o part is constant under $\mathbb{E}_{p(x^m \mid x^o)}$ and factors out:

$$\begin{aligned}
\mathbb{E}_{p(x^m \mid x^o)} [\exp(-i z u)] &= \mathbb{E}_{p(x^m \mid x^o)} \left[\exp\left(-i(w_o^T x^o + w_m^T x^m)u\right) \right] \\
&= \exp\left(-i w_o^T x^o u\right) \cdot \mathbb{E}_{p(x^m \mid x^o)} \left[\exp\left(-i w_m^T x^m u\right) \right] \\
&= \exp\left(-i w_o^T x^o u\right) \cdot \phi_{x^m \mid x^o}(-u w_m)
\end{aligned}$$

So the input to the characteristic function is $s := -uw_m$.

1.1.6 Full Formula

Plugging the result of the previous subsection back into the outer expression for $p(y \mid x^o)$, we obtain:

$$\begin{aligned} p(y \mid x^o) &= \frac{1}{2} \mathbb{E}_{\gamma \sim \text{PG}(1,0)} \mathbb{E}_{u \sim \mathcal{N}(0,\gamma)} \left[\exp \left(\frac{1 + 4iu}{8\gamma} \right) \cdot \exp \left(-iw_o^T x^o u \right) \cdot \phi_{x^m|x^o}(-uw_m) \right] \\ &= \frac{1}{2} \mathbb{E}_{\gamma \sim \text{PG}(1,0)} \mathbb{E}_{u \sim \mathcal{N}(0,\gamma)} \left[\exp \left(\frac{1}{8\gamma} + \frac{iu}{2\gamma} - iw_o^T x^o u \right) \cdot \phi_{x^m|x^o}(-uw_m) \right] \\ &= \frac{1}{2} \mathbb{E}_{\gamma \sim \text{PG}(1,0)} \mathbb{E}_{u \sim \mathcal{N}(0,\gamma)} \left[\exp \left(\frac{1}{8\gamma} + iu \left(\frac{1}{2\gamma} - w_o^T x^o \right) \right) \cdot \phi_{x^m|x^o}(-uw_m) \right] \end{aligned}$$

We propose this as a final expression that we can tractably evaluate.

1.2 Implementation

Two ingredients are still abstract in the formula above: the conditional characteristic function $\phi_{x^m|x^o}$, which depends on the density model, and the inner $\mathcal{N}(0, \gamma)$ expectation, which needs to be approximated. This section makes both concrete.

1.2.1 Conditional CF for a GMM

For a Gaussian mixture model $p(x) = \sum_{k=1}^K \pi_k \mathcal{N}(x; \mu_k, \Sigma_k)$ with diagonal covariances $\Sigma_k = \text{diag}(\sigma_k^2)$, the conditional density $p(x^m \mid x^o)$ is again a mixture, with the same component shapes restricted to the missing coordinates and reweighted by the posterior responsibility of each component given x^o :

$$\tilde{\pi}_k(x^o) = \frac{\pi_k \mathcal{N}(x^o; \mu_k^o, \Sigma_k^o)}{\sum_{j=1}^K \pi_j \mathcal{N}(x^o; \mu_j^o, \Sigma_j^o)}.$$

Since the characteristic function of $\mathcal{N}(\mu, \Sigma)$ at s is $\exp \left(is^T \mu - \frac{1}{2} s^T \Sigma s \right)$, and the CF of a mixture is the same convex combination of component CFs, we get a closed-form expression:

$$\phi_{x^m|x^o}(s) = \sum_{k=1}^K \tilde{\pi}_k(x^o) \exp \left(is^T \mu_k^m - \frac{1}{2} s^T \Sigma_k^m s \right),$$

where μ_k^m, Σ_k^m are the component means and covariances restricted to the missing coordinates.

1.2.2 Conditional CF for a General PC

More generally, any smooth and decomposable probabilistic circuit admits the conditional CF in closed form by recursive evaluation, since $\phi_X(s) = \mathbb{E}[\exp(is^T X)]$ is itself an expectation of a function that factorises over disjoint variable scopes, and PCs are by construction tractable for exactly this kind of expectation. Concretely, for a node n over scope $\text{sc}(n)$ with input s restricted to that scope:

- **Leaf node** over a single variable X with density $p_n(x)$: $\phi_n(s) = \int \exp(isx) p_n(x) dx$, which is closed-form for the standard leaf families (Gaussian, categorical, ...).
- **Product node** with children $n_1, \dots, n_C$ on disjoint scopes (decomposability): variables in different children are independent, so the CF factorises, $\phi_n(s) = \prod_{c=1}^C \phi_{n_c}(s_{\text{sc}(n_c)})$.
- **Sum node** with weights $w_c \geq 0$, $\sum_c w_c = 1$: the CF is the same convex combination of children's CFs, $\phi_n(s) = \sum_{c=1}^C w_c \phi_{n_c}(s)$.

Conditioning on x^o amounts to evaluating the observed-scope leaves at x^o and renormalising the sum weights to posterior responsibilities, in exact analogy to the GMM case above; the missing-scope evaluation then returns $\phi_{x^m|x^o}(s)$ in a single bottom-up pass.

1.2.3 Gauss–Hermite Quadrature

The inner Gaussian expectation is amenable to Gauss–Hermite quadrature, with deterministic nodes $\{t_i\}_{i=1}^N$ and weights $\{\omega_i\}_{i=1}^N$ such that

$$\int_{-\infty}^{\infty} e^{-t^2} g(t) dt \approx \sum_{i=1}^N \omega_i g(t_i)$$

To use this we need to bring the $\mathcal{N}(0, \gamma)$ expectation into the form $\int e^{-t^2} g(t) dt$. Writing the expectation out as an integral against the Gaussian density gives

$$\mathbb{E}_{u \sim \mathcal{N}(0, \gamma)}[f(u)] = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\gamma}} \exp\left(-\frac{u^2}{2\gamma}\right) f(u) du.$$

Apply the substitution $u = \sqrt{2\gamma} t$. It changes three pieces of the integrand:

- the differential: $du = \sqrt{2\gamma} dt$,

- the Gaussian exponent: $\frac{u^2}{2\gamma} = \frac{(\sqrt{2\gamma}t)^2}{2\gamma} = \frac{2\gamma t^2}{2\gamma} = t^2$,
- the argument of f : $f(u) = f(\sqrt{2\gamma}t)$,

and the integration limits are unchanged since $\sqrt{2\gamma} > 0$. Plugging all three substitutions into the integral:

$$\begin{aligned}\mathbb{E}_{u \sim \mathcal{N}(0, \gamma)}[f(u)] &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\gamma}} e^{-t^2} f(\sqrt{2\gamma}t) \sqrt{2\gamma} dt \\ &= \frac{\sqrt{2\gamma}}{\sqrt{2\pi\gamma}} \int_{-\infty}^{\infty} e^{-t^2} f(\sqrt{2\gamma}t) dt.\end{aligned}$$

The constant prefactor simplifies to

$$\frac{\sqrt{2\gamma}}{\sqrt{2\pi\gamma}} = \sqrt{\frac{2\gamma}{2\pi\gamma}} = \sqrt{\frac{1}{\pi}} = \frac{1}{\sqrt{\pi}},$$

so the Gaussian expectation becomes

$$\mathbb{E}_{u \sim \mathcal{N}(0, \gamma)}[f(u)] = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-t^2} f(\sqrt{2\gamma}t) dt.$$

Which we then can apply by

$$\mathbb{E}_{u \sim \mathcal{N}(0, \gamma)}[f(u)] \approx \frac{1}{\sqrt{\pi}} \sum_{i=1}^N \omega_i f(\sqrt{2\gamma}t_i).$$

Plugging this into the inner expectation of our final formula, with

$$f(u) = \exp\left(\frac{1}{8\gamma} + iu\left(\frac{1}{2\gamma} - w_o^T x^o\right)\right) \phi_{x^m|x^o}(-uw_m)$$

gives

$$p(y | x^o) \approx \frac{1}{2\sqrt{\pi}} \mathbb{E}_{\gamma \sim \text{PG}(1,0)} \left[\sum_{i=1}^N \omega_i \exp\left(\frac{1}{8\gamma} + i\sqrt{2\gamma}t_i\left(\frac{1}{2\gamma} - w_o^T x^o\right)\right) \phi_{x^m|x^o}(-\sqrt{2\gamma}t_i w_m) \right]$$

1.2.4 Gauss–Pólya–Gamma Quadrature

The outer expectation over $\gamma \sim \text{PG}(1,0)$ can likewise be handled deterministically by constructing a bespoke quadrature rule for the $\text{PG}(1,0)$ measure. The key ingredient is

the *Golub–Welsch algorithm* [4], which produces n -point quadrature nodes and weights from the first $2n$ moments of any positive measure.

Moments of $\text{PG}(1, 0)$. The Laplace transform of $\text{PG}(1, 0)$ is known in closed form:

$$\mathcal{L}(t) := \mathbb{E}_{\gamma \sim \text{PG}(1,0)}[e^{-t\gamma}] = \text{sech}\left(\sqrt{\frac{t}{2}}\right) = \frac{1}{\cosh\left(\sqrt{t/2}\right)}.$$

The raw moments follow from derivatives at zero:

$$\mu_k := \mathbb{E}[\gamma^k] = (-1)^k \mathcal{L}^{(k)}(0),$$

which can be evaluated numerically to arbitrary precision (we use `mpmath` at 200 decimal places to avoid catastrophic cancellation for large k).

Golub–Welsch construction. Given $\mu_0, \dots, \mu_{2n-1}$, the algorithm proceeds as follows. Form the $n \times n$ Hankel moment matrices

$$H_{ij} = \mu_{i+j}, \quad H'_{ij} = \mu_{i+j+1}, \quad i, j = 0, \dots, n-1.$$

Compute the Cholesky factor $H = R^\top R$ and the Jacobi (tridiagonal) matrix

$$J = R^{-\top} H' R^{-1}, \quad \text{symmetrised as } J \leftarrow \frac{1}{2}(J + J^\top).$$

Eigendecompose $J = V \Lambda V^\top$. Then the quadrature nodes and weights are

$$\gamma_j = \lambda_j, \quad w_j = \mu_0 v_{0j}^2, \quad j = 1, \dots, n,$$

where λ_j are the eigenvalues and v_{0j} is the first component of the j -th eigenvector. The resulting rule integrates polynomials in γ exactly up to degree $2n - 1$.

Numerical considerations. The Hankel matrix H becomes numerically singular in double precision for $n \gtrsim 12$. We therefore perform the entire Golub–Welsch computation in high-precision arithmetic (`mpmath`, `dps=200`) and convert to float 32 only at the end. The maximum reliable order is $n = 19$; beyond that, eigenvalues become negative at any tested precision.

Fully deterministic formula. Combining Gauss–PG nodes $(\gamma_j, w_j)_{j=1}^M$ with Gauss–Hermite nodes $(t_i, \omega_i)_{i=1}^N$ and applying both quadrature rules to the full double expec-

tation yields the *zero-variance* approximation

$$p(y \mid x^o) \approx \frac{1}{2\sqrt{\pi}} \sum_{j=1}^M w_j \sum_{i=1}^N \omega_i \exp\left(\frac{1}{8\gamma_j} + i\sqrt{2\gamma_j} t_i \left(\frac{1}{2\gamma_j} - w_o^T x^o\right)\right) \phi_{x^m|x^o}(-\sqrt{2\gamma_j} t_i w_m).$$

1.2.5 Algorithms

Putting the pieces together for a diagonal-covariance GMM density. Algorithm 1 uses Monte Carlo for both integrals. Algorithm 2 keeps MC for the outer PG integral but replaces the inner Gaussian MC with Gauss–Hermite quadrature, eliminating inner variance. Algorithm 3 further replaces the PG Monte Carlo with Gauss–PG quadrature, making the estimator fully deterministic.

Algorithm 1 PCPG: expected sigmoid under a GMM, full Monte Carlo

Input: LR weights w , partial observation x^o (missing entries marked), GMM parameters $\{\pi_k, \mu_k, \Sigma_k\}_{k=1}^K$, PG samples M , Gaussian samples L

- 1: Split $w = (w_o, w_m)$ and $\mu_k = (\mu_k^o, \mu_k^m)$, $\Sigma_k = (\Sigma_k^o, \Sigma_k^m)$ by the observed/missing mask
 - 2: $\tilde{\pi}_k \leftarrow \frac{\pi_k \mathcal{N}(x^o; \mu_k^o, \Sigma_k^o)}{\sum_j \pi_j \mathcal{N}(x^o; \mu_j^o, \Sigma_j^o)}$ for $k = 1, \dots, K$
 - 3: $c \leftarrow w_o^T x^o$
 - 4: $S \leftarrow 0$
 - 5: **for** $j = 1, \dots, M$ **do**
 - 6: Sample $\gamma_j \sim \text{PG}(1, 0)$
 - 7: **for** $\ell = 1, \dots, L$ **do**
 - 8: Sample $u \sim \mathcal{N}(0, \gamma_j)$
 - 9: $s \leftarrow -u w_m$
 - 10: $\phi \leftarrow \sum_{k=1}^K \tilde{\pi}_k \exp\left(is^T \mu_k^m - \frac{1}{2}s^T \Sigma_k^m s\right)$
 - 11: factor $\leftarrow \exp\left(\frac{1}{8\gamma_j} + iu \left(\frac{1}{2\gamma_j} - c\right)\right)$
 - 12: $S \leftarrow S + \text{factor} \cdot \phi$
 - 13: **end for**
 - 14: **end for**
 - 15: **return** $\frac{1}{2ML} \cdot \text{Re}(S)$
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Algorithm 2 PCPG-GH: expected sigmoid under a GMM, MC outer + Gauss–Hermite inner

Input: LR weights w , partial observation x^o (missing entries marked), GMM parameters $\{\pi_k, \mu_k, \Sigma_k\}_{k=1}^K$, PG samples M , Hermite order N

- 1: $(t_i, \omega_i)_{i=1}^N \leftarrow \text{hermgauss}(N)$ $\triangleright$ deterministic nodes/weights, precomputed once
- 2: Split $w = (w_o, w_m)$ and $\mu_k = (\mu_k^o, \mu_k^m)$, $\Sigma_k = (\Sigma_k^o, \Sigma_k^m)$ by the observed/missing mask
- 3: $\tilde{\pi}_k \leftarrow \frac{\pi_k \mathcal{N}(x^o; \mu_k^o, \Sigma_k^o)}{\sum_j \pi_j \mathcal{N}(x^o; \mu_j^o, \Sigma_j^o)}$ for $k = 1, \dots, K$
- 4: $c \leftarrow w_o^T x^o$
- 5: $S \leftarrow 0$
- 6: **for** $j = 1, \dots, M$ **do**
- 7: Sample $\gamma_j \sim \text{PG}(1, 0)$
- 8: **for** $i = 1, \dots, N$ **do**
- 9: $u \leftarrow \sqrt{2\gamma_j} t_i$
- 10: $s \leftarrow -u w_m$
- 11: $\phi \leftarrow \sum_{k=1}^K \tilde{\pi}_k \exp\left(is^T \mu_k^m - \frac{1}{2}s^T \Sigma_k^m s\right)$
- 12: factor $\leftarrow \exp\left(\frac{1}{8\gamma_j} + iu \left(\frac{1}{2\gamma_j} - c\right)\right)$
- 13: $S \leftarrow S + \omega_i \cdot \text{factor} \cdot \phi$
- 14: **end for**
- 15: **end for**
- 16: **return** $\frac{1}{2\sqrt{\pi} M} \cdot \text{Re}(S)$

Algorithm 3 PCPG-Quad: fully deterministic expected sigmoid under a GMM

Input: LR weights w , partial observation x^o (missing entries marked), GMM parameters $\{\pi_k, \mu_k, \Sigma_k\}_{k=1}^K$, PG quadrature order M ($M \leq 19$), Hermite order N

- 1: $(\gamma_j, w_j)_{j=1}^M \leftarrow \text{pg_gauss_quadrature}(M)$ $\triangleright$ Golub–Welsch, computed once at high precision; see §1.2.4
- 2: $(t_i, \omega_i)_{i=1}^N \leftarrow \text{hermgauss}(N)$
- 3: Split $w = (w_o, w_m)$ and $\mu_k = (\mu_k^o, \mu_k^m)$, $\Sigma_k = (\Sigma_k^o, \Sigma_k^m)$ by the observed/missing mask
- 4: $\tilde{\pi}_k \leftarrow \frac{\pi_k \mathcal{N}(x^o; \mu_k^o, \Sigma_k^o)}{\sum_j \pi_j \mathcal{N}(x^o; \mu_j^o, \Sigma_j^o)}$ for $k = 1, \dots, K$
- 5: $c \leftarrow w_o^T x^o$
- 6: $S \leftarrow 0$
- 7: **for** $j = 1, \dots, M$ **do**
- 8: **for** $i = 1, \dots, N$ **do**
- 9: $u \leftarrow \sqrt{2\gamma_j} t_i$
- 10: $s \leftarrow -u w_m$
- 11: $\phi \leftarrow \sum_{k=1}^K \tilde{\pi}_k \exp\left(is^T \mu_k^m - \frac{1}{2}s^T \Sigma_k^m s\right)$
- 12: $\text{factor} \leftarrow \exp\left(\frac{1}{8\gamma_j} + iu \left(\frac{1}{2\gamma_j} - c\right)\right)$
- 13: $S \leftarrow S + w_j \cdot \omega_i \cdot \text{factor} \cdot \phi$
- 14: **end for**
- 15: **end for**
- 16: **return** $\frac{1}{2\sqrt{\pi}} \cdot \text{Re}(S)$

1.3 Discussion

1.3.1 Variance of the MC Estimator

The Monte Carlo estimator draws N samples $x^{(i)} \sim p(x^m \mid x^o)$ and returns their average sigmoid value:

$$\hat{I}_{\text{MC}} = \frac{1}{N} \sum_{i=1}^N \sigma(w^T x^{(i)}).$$

This is a standard sample mean, so its variance is

$$\text{Var}(\hat{I}_{\text{MC}}) = \frac{\sigma_{\text{MC}}^2}{N}, \quad \sigma_{\text{MC}}^2 = \text{Var}_{x \sim p(x \mid x^o)}[\sigma(w^T x)].$$

Since $\sigma(w^T x) \in (0, 1)$, the variance of any $[0, 1]$ -valued random variable is at most $\frac{1}{4}$ (attained by a Bernoulli($\frac{1}{2}$) distribution), giving the tight upper bound

$$\text{Var}(\hat{I}_{\text{MC}}) \leq \frac{1}{4N}.$$

1.3.2 Variance of the PCPG Estimator

PCPG is a two-level Monte Carlo estimator. Let $h(\gamma, u)$ denote the integrand

$$h(\gamma, u) := \text{Re} \left[\exp \left(\frac{1}{8\gamma} + iu \left(\frac{1}{2\gamma} - w_o^T x^o \right) \right) \phi_{x^m|x^o}(-u w_m) \right],$$

and let $g(\gamma) := \mathbb{E}_{u \sim \mathcal{N}(0, \gamma)}[h(\gamma, u)]$ be the inner expectation. The estimator returns

$$\hat{I}_{\text{PCPG}} = \frac{1}{2} \cdot \frac{1}{n_{pg}} \sum_{j=1}^{n_{pg}} \underbrace{\frac{1}{n_{\text{gauss}}} \sum_{k=1}^{n_{\text{gauss}}} h(\gamma^{(j)}, u^{(k,j)})}_{\approx g(\gamma^{(j)})}.$$

Applying the law of total variance to the inner mean gives

$$\text{Var}(\hat{I}_{\text{PCPG}}) = \frac{1}{4} \left[\frac{\sigma_B^2}{n_{pg}} + \frac{\sigma_A^2}{n_{pg} \cdot n_{\text{gauss}}} \right],$$

where

$$\begin{aligned} \sigma_B^2 &= \text{Var}_{\gamma \sim \text{PG}(1,0)}[g(\gamma)], \quad (\text{outer PG variance, irreducible by increasing } n_{\text{gauss}}) \\ \sigma_A^2 &= \mathbb{E}_{\gamma \sim \text{PG}(1,0)} \left[\text{Var}_{u \sim \mathcal{N}(0, \gamma)}[h(\gamma, u)] \right]. \quad (\text{inner Gaussian variance}) \end{aligned}$$

Unlike the MC case, neither σ_A^2 nor σ_B^2 is bounded by $\frac{1}{4}$. The integrand $h(\gamma, u)$ contains the factor $\exp(1/(8\gamma))$, which grows without bound as $\gamma \rightarrow 0^+$; since $\text{PG}(1, 0)$ assigns positive probability to arbitrarily small γ , both variance components are in principle unbounded.

1.3.3 Variance of the PCPG-GH Estimator

PCPG-GH (Algorithm 2) replaces the inner Monte Carlo sum with Gauss–Hermite quadrature. Using N_H Hermite nodes (t_i, ω_i) , the inner integral is approximated

deterministically as

$$\tilde{g}(\gamma) := \frac{1}{\sqrt{\pi}} \sum_{i=1}^{N_H} \omega_i h(\gamma, \sqrt{2\gamma} t_i) \approx g(\gamma).$$

The estimator is then a *single-level* Monte Carlo average over the PG draws:

$$\hat{I}_{\text{PCPG-GH}} = \frac{1}{2} \cdot \frac{1}{n_{pg}} \sum_{j=1}^{n_{pg}} \tilde{g}(\gamma^{(j)}).$$

There is no inner randomness, so the law of total variance reduces immediately to

$$\text{Var}(\hat{I}_{\text{PCPG-GH}}) = \frac{1}{4} \cdot \frac{\text{Var}_{\gamma}[\tilde{g}(\gamma)]}{n_{pg}} \xrightarrow{N_H \rightarrow \infty} \frac{\sigma_B^2}{4 n_{pg}}.$$

The σ_A^2 term present in PCPG is entirely eliminated: Gauss–Hermite quadrature removes all inner variance at the cost of N_H deterministic CF evaluations per PG draw instead of n_{gauss} random ones.

Bias of PCPG-GH. Unlike the full-MC PCPG estimator, PCPG-GH is *biased* for any finite N_H because $\tilde{g}(\gamma) \neq g(\gamma)$ in general:

$$\text{Bias}(\hat{I}_{\text{PCPG-GH}}) = \frac{1}{2} \mathbb{E}_{\gamma \sim \text{PG}(1,0)}[\tilde{g}(\gamma) - g(\gamma)].$$

The inner error $\tilde{g}(\gamma) - g(\gamma)$ is the error of an N_H -point Gauss–Hermite rule applied to the integral $\int e^{-t^2} h(\gamma, \sqrt{2\gamma} t) dt$. The classical remainder formula states that for a $2N_H$ -times differentiable integrand f :

$$\int_{-\infty}^{\infty} e^{-t^2} f(t) dt - \sum_{i=1}^{N_H} \omega_i f(t_i) = \frac{\sqrt{\pi} (N_H)!}{2^{2N_H} (2N_H)!} f^{(2N_H)}(\xi)$$

for some $\xi \in \mathbb{R}$. The prefactor $\frac{(N_H)!}{2^{2N_H} (2N_H)!}$ decays super-exponentially in N_H (faster than any polynomial), so the inner bias is negligible in practice for $N_H \geq 20$: the integrand $h(\gamma, \sqrt{2\gamma} t)$ is analytic in t (it involves complex exponentials and the GMM CF, which are entire functions), so all derivatives exist and are bounded on compact sets. In summary, PCPG-GH trades a small, rapidly vanishing systematic error for the complete elimination of inner-loop variance.

1.3.4 Variance and Bias of the PCPG-Quad Estimator

PCPG-Quad (Algorithm 3) is fully deterministic: both the outer PG expectation and the inner Gaussian expectation are replaced by quadrature rules, so it has zero variance but nonzero bias from both approximations.

Variance. Since the estimator is deterministic, $\text{Var}(\hat{I}_{\text{PCPG-Quad}}) = 0$. Running it multiple times with the same inputs always returns the same value.

Bias from the inner (Gauss–Hermite) quadrature. The inner approximation $\tilde{g}(\gamma) \approx g(\gamma)$ carries the same super-exponentially vanishing error as in PCPG-GH, and is negligible for $N_H \geq 20$.

Bias from the outer (Gauss–PG) quadrature. The outer integral $\frac{1}{2} \mathbb{E}_{\gamma \sim \text{PG}(1,0)}[\tilde{g}(\gamma)]$ is replaced by the M -point quadrature sum $\frac{1}{2} \sum_{j=1}^M w_j \tilde{g}(\gamma_j)$. An M -point Gauss quadrature rule for measure p_{PG} is exact for polynomials in γ up to degree $2M - 1$. The integrand $\tilde{g}(\gamma)$ is not a polynomial: it contains the factor $e^{1/(8\gamma)}$, which has an essential singularity at $\gamma = 0$. Consequently the quadrature error does *not* vanish at a polynomial rate but is instead controlled by the regularity of $\tilde{g}$ away from the singularity and the concentration of the $\text{PG}(1,0)$ measure near zero.

The total bias is therefore the sum of the two quadrature errors:

$$\text{Bias}(\hat{I}_{\text{PCPG-Quad}}) = \underbrace{\frac{1}{2} \mathbb{E}_{\gamma}[\tilde{g}(\gamma) - g(\gamma)]}_{\text{inner (GH), negligible for } N_H \geq 20} + \underbrace{\frac{1}{2} \left(\sum_{j=1}^M w_j \tilde{g}(\gamma_j) - \mathbb{E}_{\gamma}[\tilde{g}(\gamma)] \right)}_{\text{outer (Gauss-PG), decreasing in } M}.$$

Bias–variance tradeoff. PCPG-Quad has zero variance and a bias that decreases as M and N_H increase. For a fixed evaluation budget of $M \times N_H$ CF calls, the optimal strategy is to increase N_H until the inner bias is negligible (cheaply achieved at $N_H \approx 30$) and then increase M to drive down the outer bias. The hard limit $M \leq 19$ (imposed by numerical precision in the Golub–Welsch construction; see §1.2.4) caps the accuracy of PCPG-Quad regardless of N_H . In practice, PCPG-Quad with $M = 15$ – 19 and $N_H = 30$ gives a highly accurate, zero-variance estimate that can serve as a reference value against which the stochastic estimators are validated.

1.3.5 Budget Comparison

To compare all estimators on equal footing we define a shared *total budget* N as the total number of characteristic-function evaluations: $N = n_{\text{samples}}$ for MC, $N = n_{pg} \cdot n_{\text{gauss}}$ for PCPG, and $N = n_{pg} \cdot N_H$ for PCPG-GH. For PCPG the natural equal-split is $n_{pg} = n_{\text{gauss}} = \sqrt{N}$; for PCPG-GH, fixing N_H and setting $n_{pg} = N/N_H$ gives:

$$\text{Var}(\hat{I}_{\text{PCPG-GH}}) \approx \frac{\sigma_B^2 N_H}{4N}.$$

This recovers the $1/N$ rate of MC, with constant $\sigma_B^2 N_H/4$ in place of σ_{MC}^2 . The three estimators thus behave as follows at equal budget N :

$$\text{Var}(\hat{I}_{\text{MC}}) = \frac{\sigma_{\text{MC}}^2}{N}, \quad \text{Var}(\hat{I}_{\text{PCPG}}) \sim \frac{\sigma_B^2}{4\sqrt{N}}, \quad \text{Var}(\hat{I}_{\text{PCPG-GH}}) \approx \frac{\sigma_B^2 N_H}{4N}.$$

PCPG (full MC) decays as $1/\sqrt{N}$ and is therefore the least efficient of the three. PCPG-GH and MC both decay as $1/N$; PCPG-GH is preferable to MC when $\sigma_B^2 N_H < 4\sigma_{\text{MC}}^2$, i.e. when the outer PG variance is small relative to the sigmoid variance, but in practice σ_B^2 is typically comparable to σ_{MC}^2 so MC remains more efficient per evaluation. The true advantage of the PCPG family over MC is not variance but applicability: PCPG requires only the characteristic function of $p(x^m | x^o)$, not the ability to draw conditional samples. A natural equal-split allocation for PCPG sets $n_{pg} = n_{\text{gauss}} = \sqrt{N}$, giving

$$\text{Var}(\hat{I}_{\text{PCPG}}) = \frac{1}{4} \left[\frac{\sigma_B^2}{\sqrt{N}} + \frac{\sigma_A^2}{N} \right] \underset{N \rightarrow \infty}{\sim} \frac{\sigma_B^2}{4\sqrt{N}}.$$

The dominant term decays as $1/\sqrt{N}$, whereas the MC variance decays as $1/N$. PCPG is therefore *less* variance-efficient than MC at equal budget under this allocation.

1.3.6 Values Outside $[0, 1]$

The MC estimator is always in $(0, 1)$ because σ maps into that interval. PCPG, by contrast, computes the real part of a Fourier-type integral approximated by Monte Carlo, and individual realisations are *not* constrained to $[0, 1]$: the estimator can return negative values or values greater than one when n_{pg} or n_{gauss} is small. This is not a correctness problem — the estimator is unbiased and converges to a value in $[0, 1]$ — but it is a practical nuisance when the output is interpreted as a probability.

1.3.7 When is PCPG Preferable?

The variance comparison above may suggest that PCPG offers no advantage over MC. Its value lies elsewhere: PCPG never requires samples from $p(x^m | x^o)$. It only needs the *characteristic function* of the conditional distribution, which is available in closed form for GMMs and general smooth decomposable PCs. In settings where drawing conditional samples is expensive or intractable — for instance with non-Gaussian leaf distributions or densely connected circuits — PCPG remains applicable while plain MC does not. Furthermore, when the inner integral is handled by Gauss–Hermite quadrature (the PCPG-GH variant), the inner variance σ_A^2 is eliminated entirely, reducing the variance to $\sigma_B^2/(4 n_{pg})$, which matches the $1/N$ rate of MC (with $N = n_{pg}$) up to the constant $\sigma_B^2/4$.

1.4 Related Work

Here, we just focus on works that also try to calculate

$$\mathbb{E}_{P(x^m|x^o)} \left[\sigma(w^T x) \right]$$

but use a different approach.

1.4.1 Naive Conformant Learning (NaCL)

Khosravi et al. [5] first point out that this expectation is hard to compute in general: even with a naive Bayes feature distribution, it is NP-hard. They also note that the popular shortcut of mean imputation only matches the expectation when the classifier is linear *and* the features are independent — for the sigmoid it is always biased.

Their fix is to pick a feature distribution on which the expectation *is* cheap, namely a naive Bayes model over binary features. The trick is that naive Bayes and logistic regression are dual to each other: every naive Bayes induces a logistic regression, and many naive Bayes models induce the same one. They call a naive Bayes *conformant* with the given LR if both give the same class probabilities on fully observed inputs, and learn such a distribution via geometric programming (“naive conformant learning”). Once this naive Bayes is fitted, the expected prediction is just a marginal inference query and runs in linear time.

Difference to us: they keep the feature distribution simple (naive Bayes on binary inputs) so the expectation is exact and cheap; we keep the feature distribution rich (a GMM or PC on continuous inputs) and instead use a stochastic trick to handle

the sigmoid. They also have to refit the feature distribution whenever the classifier changes, while ours is trained once and reused.

1.4.2 Joint Modelling via SAEM

Jiang, Josse, and Lavielle [6] assume a joint Gaussian model on the covariates, $x \sim \mathcal{N}(\mu, \Sigma)$, together with logistic regression on top. Their main contribution is on the training side: they fit (μ, Σ, β) jointly from training data that may itself contain missing values, using a stochastic-approximation EM algorithm with Metropolis–Hastings updates.

For prediction at test time, they simply do plain Monte Carlo: sample $x^{m,(s)} \sim p(x^m \mid x^o)$ from the fitted Gaussian and average,

$$p(y \mid x^o) \approx \frac{1}{S} \sum_{s=1}^S p(y \mid x^o, x^{m,(s)})$$

This is exactly the MC baseline we compare against, just with a single Gaussian instead of a mixture as the density.

Difference to us: they sample the missing features and average the sigmoid; we never sample the missing features, but instead rewrite the sigmoid so that only the characteristic function of $p(x^m \mid x^o)$ is needed. Their feature distribution is also limited to a single Gaussian, while ours can capture multimodal shapes.

1.5 Why Monte Carlo Alone Is Not Enough

Monte Carlo estimation of $p(y \mid x^o)$ — drawing N samples $x^{m,(i)} \sim p(x^m \mid x^o)$ and averaging the sigmoid predictions — is unbiased and simple. However, it has two fundamental limitations.

1.5.1 Conditional Sampling May Be Intractable

MC requires the ability to *draw samples* from $p(x^m \mid x^o)$. For a GMM with diagonal covariances this is straightforward, but for a general smooth decomposable PC it is not always tractable. Computing the conditional *probability* $p(x^o)$ takes linear time in the circuit size, but drawing a sample from $p(x^m \mid x^o)$ requires a top-down ancestral pass that branches on sum-node weights recomputed with the evidence. For more expressive circuit architectures this pass may be infeasible.

By contrast, the *characteristic function* $\phi_{x^m|x^o}(s)$ is always tractable for smooth decomposable PCs: it is an expectation of a function that factorises over disjoint variable scopes, which PCs are built to handle efficiently (see §??). PCPG therefore applies wherever MC does not.

1.5.2 No Closed Form Exists for the Sigmoid Integral

Even for the simplest non-trivial density — a single Gaussian $p(x^m | x^o) = \mathcal{N}(\mu, \Sigma)$ — the integral

$$\mathbb{E}[\sigma(w^T x) | x^o] = \int_{-\infty}^{\infty} \sigma(\mu_w + \sigma_w z) \frac{e^{-z^2/2}}{\sqrt{2\pi}} dz, \quad \mu_w = w_m^T \mu, \quad \sigma_w^2 = w_m^T \Sigma w_m,$$

has no closed form. Approximations exist — the probit surrogate $\sigma(z) \approx \Phi(\sqrt{\pi/8} z)$ yields $\Phi(\mu_w / \sqrt{1 + \pi \sigma_w^2 / 8})$ — but they introduce systematic errors. For a GMM the integral must be approximated separately for each mixture component. This means MC (or a PCPG-style estimator) is unavoidable whenever an accurate, unbiased answer is required.

1.5.3 Summary

Table 1.1 compares MC and PCPG directly.

Table 1.1: MC vs. PCPG for computing $\mathbb{E}[\sigma(w^T x) | x^o]$.

	Monte Carlo	PCPG (ours)
Unbiased	Yes	Yes (in limit)
Closed form	No	No
Requires CF of $p(x^m x^o)$	No	Yes
Requires sampling from $p(x^m x^o)$	Yes	No
Variance	σ_{MC}^2 / N	$\sigma_B^2 / (4n_{pg})^*$
Applicable to general PCs	Limited	Yes

*PCPG-GH variant; full-MC PCPG has an additional inner variance term.

Appendix

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